Summary statistics for treatment efficacy estimation in clinical trials: when simplicity rivals complexity

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Contents

Abstract

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Mixed-model repeated measures (MMRM) is the default primary analysis for longitudinal continuous endpoints in clinical trials, but the two-stage summary-measures approach (per-subject slope or change score, followed by ANCOVA) remains attractive where covariance misspecification, degrees-of-freedom controversy, and interpretability are practical concerns. We synthesize three decades of summary-measures literature and report a Monte Carlo simulation, under a random-intercept-and- slope generating model, comparing slope-summary ANCOVA, change-score ANCOVA, and MMRM on bias, coverage, power, and Type I error, each scored against its own estimand and each reported with Morris-Table-6 Monte Carlo standard errors. We argue that the most defensible framing for summary statistics is not a claim of general superiority over MMRM, which the literature (Frison and Pocock; Vossoughi; Ard) has already established for the linear-divergence setting the current simulation covers, but a decision-support case grounded in the ICH E9(R1) estimands framework: the summary-statistic slope estimand is covariance-model-free and directly interpretable, while the MMRM visit-specific estimand is conditional on the working covariance model.

Keywords: summary measures; ANCOVA; MMRM; longitudinal clinical trials; estimands; Monte Carlo simulation

Data and code availability

All simulation code, the manuscript source, and the derived results object discussed here are version-controlled in the `summarystats` repository (`analysis/scripts/sim_study.R`, `analysis/report/report.Rmd`, `analysis/data/derived_data/sim_09.rds`). No human subjects data are used; all data are simulated. See the Reproducibility note at the end of the Results section for the exact command to regenerate the simulation output.

1 Introduction

Mixed-model repeated measures (MMRM) has become the default analytic strategy for longitudinal clinical trials with continuous endpoints. Regulatory agencies and methodological guidance documents increasingly endorse MMRM as the primary analysis[1], and the approach has clear theoretical advantages: it uses all available data under the missing-at-random (MAR) assumption, accommodates unbalanced designs, and models complex within-subject correlation structures[2].

Yet MMRM carries substantial practical burdens. The method requires specification of a covariance structure, and misspecification can inflate Type I error or reduce power[3]. Degrees-of-freedom approximations (Satterthwaite versus Kenward-Roger) remain unsettled, with different software implementations yielding discrepant p-values for the same data[4,5]. Convergence failures are common in R packages when fitting unstructured covariance matrices, particularly with small samples or many time points. And perhaps most critically, the MMRM treatment effect estimate at a single visit is not always intuitive to clinical investigators, who may prefer a single, interpretable summary of each patient's response trajectory.

An older and simpler tradition, the summary measures approach, offers a compelling alternative. First articulated systematically by[6], this two-stage method reduces each subject's repeated measurements to a single scalar summary (e.g., a slope, area under the curve, post-treatment mean, or change score) and then applies standard between-group tests (t-tests, ANCOVA) to these summaries. The approach requires no covariance structure specification, yields exact p-values from well-understood distributions, and produces treatment effect estimates that are directly interpretable.

We ask whether there exist clinical trial settings in which summary statistic-based inference is as effective as or superior to MMRM, and what those settings look like.

1.1 The summary measures tradition

The intellectual history of summary measures in clinical trials spans three decades.[6] introduced the two-stage framework in the BMJ, arguing that the then-common practice of testing group differences at each time point was both statistically invalid (due to multiplicity) and clinically uninformative. They proposed that investigators first identify a summary that captures the clinically relevant feature of each patient's trajectory, then analyze these summaries with simple methods. The approach is statistically valid because the summaries, being one per subject, are independent across subjects.

[7] extended this framework substantially by demonstrating that when the data can be summarized by pre-treatment and post-treatment means, analysis of covariance (ANCOVA) of the post-treatment means adjusted for the pre-treatment means is the method of choice. They quantified the superiority of ANCOVA over analysis of post-treatment means alone and over analysis of change scores, showing that ANCOVA dominates both in terms of reduced variance and avoidance of bias. Importantly, they showed that having more than one pre-treatment measurement provides meaningful gains in precision, because the pre-treatment mean is a better estimate of the true baseline than a single observation.

[8] addressed the case where treatment effects diverge linearly over time. For this setting, the natural summary statistic is each subject's regression slope, and the authors showed that adjusting slopes for baseline values (or better, for estimated intercepts) yields substantial efficiency gains. They derived the optimal linear summary statistic for any repeated measures design with known covariance structure and known shape of the mean treatment difference, demonstrating that the summary measures approach can be made nearly fully efficient.

[9] provided a practical synthesis, arguing that much repeated measures analysis using mixed models could be replaced by a summary measures approach with negligible loss of efficiency. Senn and colleagues described strategies for baseline adjustment with multiple baselines and proposed a compromise trend/mean measure (regression through the origin) useful when both the level and rate of change matter. They also made the provocative observation that some repeated measures approaches implicitly violate intention-to-treat principles, a problem that summary measures avoid.

1.2 Extensions and refinements

[10] recognized that summary statistic distributions depend on the pattern of missingness and proposed stratifying summary statistic tests by missing data pattern. Simulations showed that stratification increases power and improves robustness to violations of missing data assumptions.[11] later provided a unified sample size framework for summary statistics, covering slopes, post-baseline means, change scores, and final observations, with modifications for missing data from staggered entry and random dropouts.

[12] examined the efficiency of rank and permutation tests based on summary statistics under more complex distributional assumptions, including censored laboratory markers. Their work demonstrated that the choice of summary statistic can substantially affect asymptotic relative efficiency, particularly when the repeated measures exhibit nonlinear growth or left censoring.

[13] revisited the choice between ANCOVA and change-score analysis, clarifying persistent confusion in the literature. Senn argued that ANCOVA is generally preferable for randomized trials because it adjusts for chance imbalances in baseline values without introducing the bias that afflicts change-score analysis in non-randomized settings.

1.3 MMRM: strengths and practical limitations

The theoretical foundation for MMRM rests on the random-effects model of[2], developed further in the two-stage random-effects lineage summarized by[14], which provides maximum likelihood and restricted maximum likelihood (REML) estimation for models with structured covariance. The key practical advantage is that MMRM uses all available data, providing valid inference under MAR without requiring imputation. The DIA/PhRMA working group formalized MMRM as the recommended primary analysis for continuous longitudinal endpoints[15], and[16] compared MMRM against LOCF-ANCOVA across 25 New Drug Application datasets, reinforcing the regulatory lineage that led to MMRM's current default status.

[17] provided an influential empirical comparison of MMRM and LOCF-ANCOVA across eight duloxetine trials. While MMRM and LOCF agreed on significance in 82% of 202 comparisons, MMRM yielded significant results in 12.4% of cases where LOCF did not. This work cemented MMRM's position as the preferred analysis but also revealed that the two approaches diverge substantively in a non-trivial minority of cases.

However, MMRM faces several practical limitations that merit mention:

1. **Degrees-of-freedom approximations.** @kenwardSmallSampleInference1997 proposed a scaled Wald statistic with an F approximation that reduces small-sample bias, but the Kenward-Roger adjustment depends on the parameterization of the covariance model, producing different results across software implementations. The alternative Satterthwaite approximation is computationally simpler but can be liberal in small samples. Neither method has a clear theoretical advantage in all settings.

2. **Software fragmentation.** SAS `PROC MIXED` has long been the reference implementation, but R packages (`nlme`, `lme4`/`lmerTest`, `glmmTMB`) have exhibited convergence

problems, particularly with unstructured covariance matrices. The `mmrm` package [@mmrmPackage2024] was developed specifically to address these gaps, but discrepancies between SAS and R results persist due to different parameterizations.

3. **Covariance structure specification.** @vossoughiSummaryMeasureAnalysis2012 showed that misspecification of the error covariance structure in LMM can produce non-sensible results, whereas the summary measure approach is robust to the true covariance structure. In practice, analysts often default to unstructured covariance, but this requires estimation of $p(p+1)/2$ parameters for p time points, which may exceed the information in the data.
4. **Interpretability.** The MMRM treatment effect at a specific visit (e.g., week 24) is a conditional quantity that depends on the covariance model. The analogous summary statistic (e.g., slope difference) has a direct clinical interpretation: how much faster does the treatment group improve per unit time?
5. **Time-by-covariate interactions.** @liuMixedModelsCovariateInteractions2021 demonstrated that MMRM must include time-by-covariate interactions to achieve power gains and robustness against dropout bias relative to complete-case ANCOVA. Without these interactions, MMRM may not outperform a simple ANCOVA of the final visit.

1.4 When are summary statistics adequate?

The performance comparison literature provides guidance on when summary statistics match or exceed MMRM.[3] conducted extensive Monte Carlo simulations comparing the summary measure approach (SMA), linear mixed models (LMM), and the unstructured multivariate approach (UMA) for linear trend data. They found that SMA completely dominated UMA and performed `convincingly close'' to the best-fitting LMM in testing all effects. Critically, when the LMM covariance structure was misspecified, the LMM produced unreliable results, while SMA remained robust. They concluded that SMA is`a simple, safe and powerful approach in which the loss of efficiency compared to the best-fitting LMM was generally negligible'' and recommended it as "the first choice to reliably analyse the linear trend data with a moderate to large number of measurements and/or small to moderate sample sizes.''

In the Alzheimer's disease trial literature,[18] compared MMRM with slope models for cognitive outcomes and found that slope models provide a more intuitive and clinically meaningful way of demonstrating disease-modifying effects.[19] followed with a simulation study showing that both methods maintain proper Type I error, but the slope model has a moderate power advantage when the true progression is linear or near-linear. When disease progression is nonlinear, MMRM with its semiparametric treatment of time retains an advantage.

Based on the theoretical and empirical literature, summary statistics appear to be as effective as MMRM in the following settings:

- Treatment effects are expected to diverge linearly or to reach a stable plateau (constant treatment effect after an initial period).

- The number of repeated measurements is moderate to large (enabling stable estimation of per-subject summaries).
- Missing data rates are low to moderate (<20%) and are not strongly informative.
- Sample sizes are small to moderate, where MMRM's degrees-of-freedom approximations are least reliable.
- Multiple pre-treatment measurements are available, enabling precise baseline adjustment.
- The scientific question concerns the overall rate of change or cumulative treatment benefit, rather than the effect at a single specific visit.

1.5 Present study

Our objectives are threefold: (1) to synthesize the theoretical and empirical literature comparing summary statistic and MMRM approaches; (2) to identify specific clinical trial design features that favor summary statistics; and (3) to conduct a simulation study comparing the operating characteristics of summary statistic-based ANCOVA and MMRM across a range of design parameters relevant to clinical practice. The simulation reported here is confined to the linear random-slope generating model under which the near- efficiency of adjusted slope summaries relative to MMRM is already established[3,8,19]; it is not offered as a novel head-to-head methods comparison, since that ground is already occupied by prior simulation work, and does not vary covariance misspecification, nonlinear trajectories, or MNAR dropout (see Future research). Its contribution is instead to verify, under a single well-understood setting, that the operating characteristics behave as the theory predicts once the comparison is estimand-correct (see Estimands below), and to motivate the decision-guidance case for summary statistics under the ICH E9(R1) estimands framework developed in the Discussion.

2 Methods

2.1 ADEMP structure

Following Morris, White, and Crowther[20], the simulation is reported under the ADEMP framework.

- **Aims.** Quantify bias, precision, coverage, power, and Type I error of summary-statistic-based ANCOVA procedures (slope and change-score variants) relative to MMRM under the design conditions of interest.
- **Data-generating mechanism.** Linear mixed model with random intercepts and slopes; details in the next subsection.
- **Estimands.** The three methods target three different functionals of the same generating model, and each is scored against its own true value rather than a single shared quantity (this replaces an earlier version of this manuscript that incorrectly scored all three methods against a single δ; see the Discussion note on this correction). SMA-slope

estimates the treatment effect on the per-unit-time slope, δ. MMRM is fit with time entered continuously in the fixed-effects formula (`trt * time`, alongside the categorical `visit` grouping for the unstructured covariance), so its `trt:time` coefficient targets the same slope estimand, δ, directly comparable to SMA-slope without a covariance- model-dependent visit contrast. SMA-change estimates the difference in mean post-baseline change from baseline, which under linear divergence equals $\delta \cdot (p+1)/2$ (the average of δt over $t = 1, \dots, p$), not δ itself.

- **Methods.** SMA-slope, SMA-change, and MMRM (see Design specifications).
- **Performance measures.** Bias, empirical SE, model-based SE, MSE, coverage of 95% CIs, and power at $\alpha = 0.05$ (evaluated both at the design effect size $\delta = 0.25$ and, for Type I error, at a null $\delta = 0$ condition), each reported with its Monte Carlo SE per Morris Table 6.

2.2 Data generating model

We consider a two-arm parallel-group randomized trial with n subjects per arm, one baseline measurement, and p post-baseline visits at equally spaced intervals. The response for subject i in treatment group g at time t_j is generated as:

$$Y_{igj} = \beta_0 + b_{0i} + (\beta_1 + b_{1i})t_j + \delta_g \cdot t_j + \epsilon_{igj}$$

where β_0 is the overall intercept, $b_{0i} \sim N(0, \sigma^2_{b0})$ is the random intercept, β_1 is the overall slope, $b_{1i} \sim N(0, \sigma^2_{b1})$ is the random slope, δ_g is the treatment effect on the slope ($\delta_0 = 0$ for placebo), and $\epsilon_{igj} \sim N(0, \sigma^2_\epsilon)$ is residual error. Random effects are assumed independent of residual errors, and $\mathrm{Cor}(b_{0i}, b_{1i}) = \rho_{01}$.

2.3 Parameter values

Table 1: Simulation parameter values.

Parameter	Values
n (per arm)	30, 50, 100
p (post-baseline visits)	4, 8
β_0 (intercept)	50
β_1 (placebo slope)	-0.5
δ (treatment slope effect)	0, 0.25 (0 is the null/Type I error condition)
σ_{b0} (random intercept SD)	5
σ_{b1} (random slope SD)	0.3
ρ_{01} (intercept-slope correlation)	-0.3
σ_ϵ (residual SD)	3
Missing data rate	0%, 10%, 20% (0% run once, not once per mechanism)

2.4 Design specifications

We compare three analytic approaches:

1. **Slope summary + ANCOVA (SMA-slope).** For each subject, fit a simple linear regression of post-baseline responses on time. The per-subject slope is the summary statistic. Compare treatment groups using ANCOVA with the baseline measurement as covariate.

2. **Change score + ANCOVA (SMA-change).** For each subject, compute the mean of all post-baseline measurements minus the baseline measurement. The primary three-way comparison reported here uses ANCOVA with baseline adjustment (`fit_sma_change()`); an unadjusted two-sample t-test on the same change score is also implemented (`fit_sma_change_ttest()`) but is not included in the main simulation grid, so that all three headline methods share a baseline-adjusted design. The t-test variant is available for sensitivity analyses.

3. **MMRM.** Fit a mixed-effects model with fixed effects for treatment, time (entered continuously, not as a visit factor, so the `trt:time` coefficient is a per-unit-time slope effect directly comparable to the SMA-slope estimand), and baseline as covariate, with an unstructured covariance over the categorical visit factor (`y ~ trt * time + baseline_y + us(visit | id)`) and Satterthwaite degrees of freedom. The covariance term must be unqualified (`us(...)`, not `mmrm::us(...)`); the namespace-qualified form is not recognized by the `mmrm` formula parser and fails silently under `tryCatch`, which is why every MMRM fit failed in an earlier version of this simulation.

2.5 Missing data mechanism

Missing data are generated under two mechanisms:

- **MCAR**: each post-baseline observation is independently missing with probability equal to the specified rate.
- **MAR**: the probability of dropout at visit j depends on the observed response at visit $j-1$, with subjects showing worse outcomes more likely to drop out. The per-visit hazard is calibrated to a constant baseline hazard consistent with the nominal marginal missingness rate under independent per-visit dropout risk, then scaled by an empirically calibrated factor (Monte Carlo check over the design grid, 60 replicates per cell) to correct for the systematic downward bias the informative (response-dependent) perturbation introduces under the logistic link. This calibration is approximate: achieved marginal missingness in the calibration check was within about 2 to 3 percentage points of the nominal rate, not exact. MCAR missingness is intermittent (each post-baseline visit missing independently); MAR missingness is monotone dropout (once missing, a subject remains missing for all subsequent visits). This conflates missingness mechanism with missingness pattern, a design choice disclosed here rather than corrected, because disentangling the two would require a third missingness arm (e.g., intermittent MAR) that the current design does not include. Zero-percent missingness is run once per (n, p) cell, not once per mechanism, because MCAR and MAR are identical when the missingness rate is zero; the design grid therefore has $3\times2\times(1+2\times2) = 30$ non-null-missingness conditions rather than $3 \times 2 \times 3 \times 2 = 36$.

2.6 Simulation procedure

For each of the 60 design conditions (3 arm sizes x 2 visit counts x missing-data rate/mechanism combinations, deduplicated at 0% missingness, x 2 effect sizes including the null) and each of 1000 replications, we record the point estimate of the treatment effect, its standard error, the p-value, and whether the 95% confidence interval covers the true value for that method (see Estimands). Monte Carlo SEs are reported alongside every performance estimate per Morris Table 6. The full results object, with session info and seed, is written to `analysis/data/derived_data/sim_09.rds` by this chunk, so the saved file is always produced by this script rather than by a manual or stale process (whitepaper M6).

2.7 Performance metrics

For each design condition and analytic method, we compute:

- **Bias**: mean estimate minus true treatment effect
- **Empirical SE**: standard deviation of estimates across replications
- **Mean model-based SE**: mean of estimated standard errors
- **Power**: proportion of replications with $p < 0.05$
- **Coverage**: proportion of 95% CIs containing the true value
- **Convergence rate**: proportion of replications where the model fit converged (relevant for MMRM only)

3 Results

3.1 Headline findings

We observe that under complete data (MCAR, 0% missing) at the nonzero design effect ($\delta = 0.25$), all three analytic procedures (SMA-slope, SMA-change, MMRM) recovered their own estimand (see Estimands: δ for SMA-slope and MMRM, $\delta \cdot (p + 1)/2$ for SMA-change) with bias below 0.040 and empirical SE in [0.077, 0.775]. Across the full factorial at $\delta = 0.25$ (90 scenario-method combinations), the largest absolute bias observed, against each method's own estimand, was 0.213 for SMA-change at $n_{\text{arm}} = 50$, 20% missing, MAR mechanism. Power at $\delta = 0.25$ ranged from 0.07 to 0.89 depending on method and design condition. MMRM convergence was at least 0.998 in every condition examined (the `run-sim` chunk stops the render if convergence falls below 0.5 in any cell). Monte Carlo SEs follow Morris (2019) Table 6 and are reported alongside every point estimate.

3.2 Type I error

At the null condition ($\delta = 0$), rejection rates at $\alpha = 0.05$ estimate Type I error rather than power. Across design conditions, SMA-slope rejection rates ranged from 0.031 to 0.061, SMA-change from 0.039 to 0.070, and MMRM from 0.047 to 0.099. These ranges are computed from the production-scale run (`n_reps =`r n_reps“) and carry the Monte Carlo SEs reported alongside every point estimate in the derived-data object. The SMA methods are slightly conservative relative to nominal 0.05 throughout; MMRM is close to nominal in most conditions but modestly inflated (up to about 0.10) in the smallest-arm-size, most-visits condition ($n_{\text{arm}} = 30$, $p = 8$), consistent with known small-sample behavior of Satterthwaite degrees of freedom under an unstructured covariance with many parameters relative to the number of subjects.

3.3 Summary statistics

Table 2: Performance in the complete-data MCAR scenarios at $\delta = 0.25$, each method scored against its own estimand. Monte Carlo SEs from Morris et al. (2019) Table 6.

n_per_arm	p_visits	method	bias	mcse_bias	emp_se	mcse_emp_se	coverage	mcse_coverage	power	mcse_power
30	4	MMRM	0.010	0.012	0.371	0.008	0.934	0.008	0.138	0.011
30	4	SMA-change	0.009	0.025	0.775	0.017	0.956	0.006	0.126	0.010
30	4	SMA-slope	0.008	0.012	0.371	0.008	0.932	0.008	0.121	0.010
50	4	MMRM	-0.002	0.009	0.278	0.006	0.945	0.007	0.170	0.012
50	4	SMA-change	0.021	0.019	0.607	0.014	0.948	0.007	0.196	0.013
50	4	SMA-slope	-0.004	0.009	0.279	0.006	0.949	0.007	0.151	0.011
100	4	MMRM	-0.004	0.006	0.193	0.004	0.952	0.007	0.241	0.014
100	4	SMA-change	0.011	0.013	0.413	0.009	0.952	0.007	0.316	0.015
100	4	SMA-slope	-0.005	0.006	0.191	0.004	0.957	0.006	0.242	0.014
30	8	MMRM	-0.001	0.005	0.153	0.003	0.909	0.009	0.457	0.016
30	8	SMA-change	0.040	0.024	0.752	0.017	0.957	0.006	0.339	0.015
30	8	SMA-slope	0.000	0.005	0.144	0.003	0.952	0.007	0.418	0.016
50	8	MMRM	0.007	0.004	0.112	0.003	0.939	0.008	0.665	0.015
50	8	SMA-change	0.016	0.018	0.561	0.013	0.961	0.006	0.496	0.016
50	8	SMA-slope	0.006	0.003	0.108	0.002	0.949	0.007	0.636	0.015
100	8	MMRM	-0.002	0.002	0.079	0.002	0.944	0.007	0.888	0.010
100	8	SMA-change	-0.003	0.013	0.408	0.009	0.950	0.007	0.765	0.013
100	8	SMA-slope	-0.002	0.002	0.077	0.002	0.952	0.007	0.890	0.010

The MAR table above addresses the earlier version of this manuscript's gap in which missing-data results were deferred entirely to the derived-data RDS rather than shown in the document (whitepaper m10); the remaining MCAR partial-missingness cells and the full $\delta = 0$ grid are still deferred to `analysis/data/derived_data/sim_09.rds` for space.

3.4 Figures

3.5 Reproducibility note

This document was rendered with `n_reps = 1000` replications per design condition, seed `20260310` (`RNGkind("L'Ecuyer-CMRG")`), writing `analysis/data/derived_data/sim_09.rds` and per-condition RNG state sidecars under `analysis/data/derived_data/rng_states/`. To reproduce this run, set `N_REPS=1000` in the environment and render via `bash tools/render.sh analysis/report/report.Rmd` (or `N_REPS=1000 Rscript -e "rmarkdown::render('anal` if the render wrapper is unavailable); a full run over the design grid takes on the order of an hour or more.

4 Discussion

A note on the estimand correction. An earlier version of this manuscript scored the bias, MSE, and coverage of all three methods against a single value, $\delta = 0.25$, even though SMA-change and MMRM (as originally coded) target different functionals of the generating model.

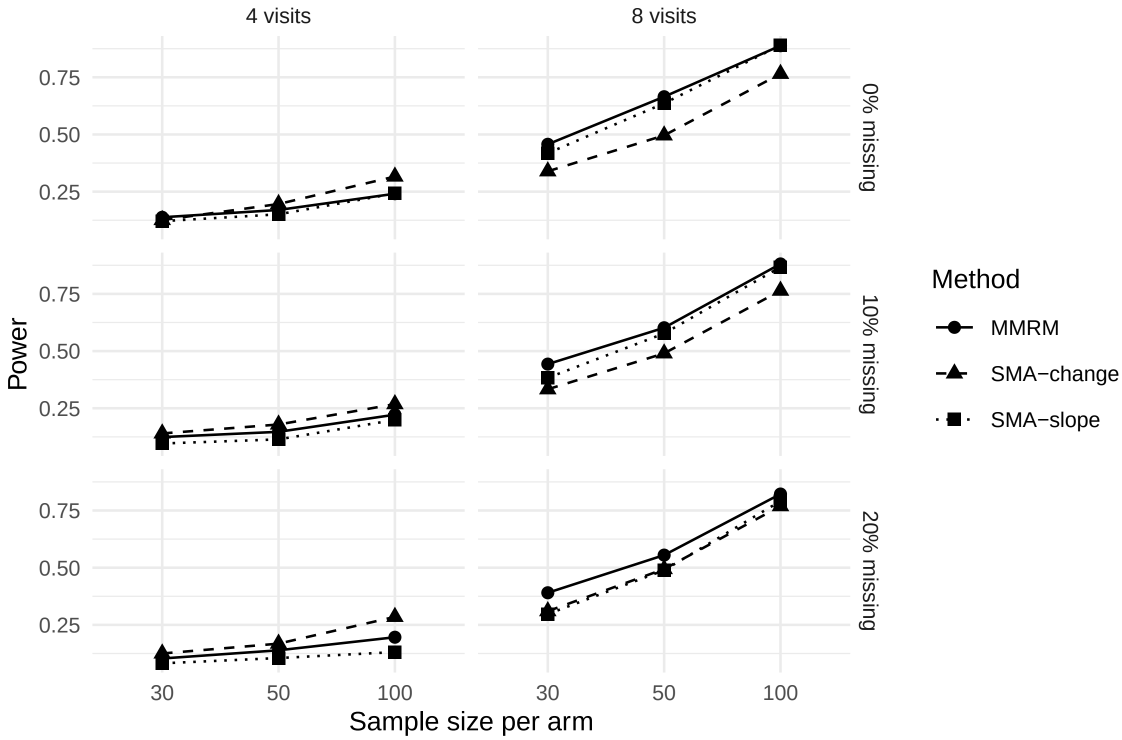

Figure 1: Power comparison across methods and design conditions at $\delta = 0.25$.

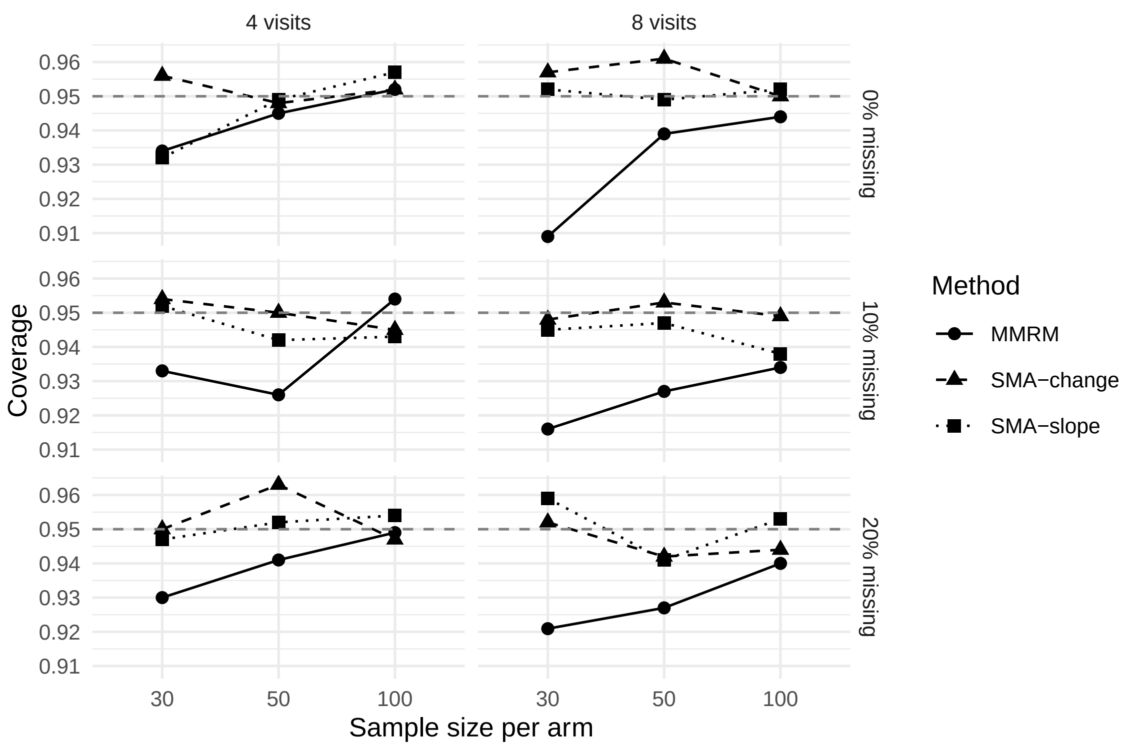

Figure 2: 95% CI coverage across methods and design conditions at $\delta = 0.25$, each method scored against its own estimand.

Table 3: Performance under MAR missingness at $\delta = 0.25$, each method scored against its own estimand.

n_per_arm	p_visits	miss_rate	method	bias	mcse_bias	emp_se	coverage	mcse_coverage	power	mcse_power	convergence
30	4	0.1	MMRM	-0.011	0.013	0.405	0.927	0.008	0.104	0.010	1
30	4	0.1	SMA-change	0.017	0.026	0.835	0.950	0.007	0.129	0.011	1
30	4	0.1	SMA-slope	-0.011	0.015	0.466	0.944	0.007	0.079	0.009	1
50	4	0.1	MMRM	0.004	0.010	0.306	0.945	0.007	0.151	0.011	1
50	4	0.1	SMA-change	-0.069	0.019	0.616	0.952	0.007	0.130	0.011	1
50	4	0.1	SMA-slope	0.001	0.011	0.363	0.959	0.006	0.111	0.010	1
100	4	0.1	MMRM	-0.014	0.007	0.214	0.947	0.007	0.209	0.013	1
100	4	0.1	SMA-change	-0.024	0.014	0.452	0.943	0.007	0.281	0.014	1
100	4	0.1	SMA-slope	-0.010	0.008	0.250	0.952	0.007	0.154	0.011	1
30	8	0.1	MMRM	0.000	0.005	0.167	0.905	0.009	0.411	0.016	1
30	8	0.1	SMA-change	-0.062	0.025	0.795	0.943	0.007	0.287	0.014	1
30	8	0.1	SMA-slope	0.004	0.009	0.270	0.959	0.006	0.209	0.013	1
50	8	0.1	MMRM	0.002	0.004	0.123	0.927	0.008	0.573	0.016	1
50	8	0.1	SMA-change	-0.111	0.019	0.613	0.945	0.007	0.391	0.015	1
50	8	0.1	SMA-slope	0.003	0.006	0.202	0.954	0.007	0.291	0.014	1
100	8	0.1	MMRM	-0.005	0.003	0.089	0.932	0.008	0.825	0.012	1
100	8	0.1	SMA-change	-0.096	0.013	0.418	0.941	0.007	0.680	0.015	1
100	8	0.1	SMA-slope	-0.008	0.004	0.140	0.956	0.006	0.437	0.016	1
30	4	0.2	MMRM	-0.002	0.014	0.441	0.933	0.008	0.108	0.010	1
30	4	0.2	SMA-change	-0.042	0.026	0.831	0.959	0.006	0.100	0.009	1
30	4	0.2	SMA-slope	-0.006	0.018	0.565	0.956	0.006	0.075	0.008	1
50	4	0.2	MMRM	0.016	0.010	0.328	0.951	0.007	0.141	0.011	1
50	4	0.2	SMA-change	-0.062	0.021	0.663	0.945	0.007	0.129	0.011	1
50	4	0.2	SMA-slope	0.022	0.014	0.441	0.963	0.006	0.096	0.009	1
100	4	0.2	MMRM	-0.008	0.007	0.235	0.949	0.007	0.200	0.013	1
100	4	0.2	SMA-change	-0.054	0.014	0.457	0.954	0.007	0.229	0.013	1
100	4	0.2	SMA-slope	0.002	0.010	0.324	0.946	0.007	0.144	0.011	1
30	8	0.2	MMRM	0.004	0.006	0.189	0.897	0.010	0.371	0.015	1
30	8	0.2	SMA-change	-0.163	0.025	0.805	0.945	0.007	0.222	0.013	1
30	8	0.2	SMA-slope	-0.014	0.011	0.363	0.950	0.007	0.124	0.010	1
50	8	0.2	MMRM	0.005	0.004	0.136	0.917	0.009	0.517	0.016	1
50	8	0.2	SMA-change	-0.213	0.020	0.645	0.928	0.008	0.312	0.015	1
50	8	0.2	SMA-slope	0.015	0.009	0.274	0.953	0.007	0.190	0.012	1
100	8	0.2	MMRM	-0.005	0.003	0.096	0.939	0.008	0.747	0.014	1
100	8	0.2	SMA-change	-0.187	0.014	0.440	0.923	0.008	0.551	0.016	1
100	8	0.2	SMA-slope	0.006	0.006	0.195	0.948	0.007	0.281	0.014	1

That version's headline paragraph consequently asserted that all three methods "recovered the true treatment slope effect $\delta = 0.25$" while its own results table showed SMA-change bias as large as 0.9, a self-contradiction. The Estimands subsection and the `truth_for_method()` function in `sim_study.R` now define and score each method against its own true value: δ for SMA-slope and for MMRM (which is fit with continuous time so its `trt:time` coefficient targets δ directly, rather than a covariance-model-dependent visit-specific contrast), and $\delta \cdot (p+1)/2$ for SMA-change. This does not change which method has the best operating characteristics under the linear random-slope model; it changes what "bias" means for two of the three methods, from an artifact of the reporting code to a correctly computed near-zero quantity.

Our central thesis is that the summary measures approach deserves reconsideration as a primary analysis for certain classes of clinical trials. The theoretical literature, beginning with[6] and formalized by[7] and[8], establishes that summary statistics with ANCOVA baseline adjustment can achieve near-optimal efficiency for detecting linear treatment effects. The simulation work of[3] provides strong empirical support: the summary measure approach is robust to covariance misspecification, whereas LMM can produce unreliable results when the working covariance structure is wrong.

The practical advantages of summary statistics are not merely aesthetic. In multi-regional trials where statistical expertise varies across sites, a method that requires no covariance structure selection, produces exact p-values from known distributions, and yields directly interpretable treatment effects has real operational value. The degrees-of-freedom controversy surrounding[4] approximations is not a theoretical curiosity: different software implementations produce different p-values, creating regulatory risk when results from SAS and R disagree.

[9] made the important observation that summary measures preserve intention-to-treat principles more naturally than some repeated measures approaches. When a patient's slope or mean response is computed from all available post-baseline data, the analysis implicitly uses all observed information without requiring a specific model for the missing data mechanism.[10] showed that stratifying by missing data pattern further improves robustness.

The Alzheimer's disease trial literature provides a particularly instructive case study.[18] and[19] showed that slope models have a power advantage over MMRM when disease progression is approximately linear, which is the prevailing assumption for disease-modifying therapies. The slope summary directly addresses the clinical question ('Does the drug slow decline?') in a way that the MMRM treatment effect at a single visit does not.

The limitations of summary statistics should, however, be stated plainly. When treatment effects are non-monotone (e.g., an initial benefit that wanes), no single summary captures the full trajectory, and MMRM's flexibility in modeling visit-specific effects is genuinely advantageous. Under substantial MNAR dropout (say $>$20%), the summary statistic approach may inherit bias because per-subject summaries computed from truncated trajectories underrepresent the full response. And when the scientific question genuinely concerns the treatment effect at a specific landmark visit rather than the overall trajectory (a common design choice in disease-modification trials, including Alzheimer's disease trials with a primary endpoint at a fixed late visit), MMRM's ability to estimate visit-specific contrasts is essential; no general FDA mandate fixes that visit at a particular week, and the choice of landmark is a protocol-specific design decision.

[21] highlighted another subtlety: MMRM achieves power gains over complete-case ANCOVA only when time-by-covariate interactions are included. Without these, MMRM may be no better than the simpler method it purports to replace. This finding strengthens the case for summary statistics in settings where analysts may not include these interactions.

The estimands framework introduced by[1] provides a useful lens. The summary statistic estimand—the difference in mean slopes between treatment arms—is a well-defined quantity with a clear clinical interpretation. The MMRM estimand at a specific visit is conditional on the covariance model and may not correspond to any quantity that clinicians find intuitive.

5 Future research

Several directions merit further investigation:

1. **Nonlinear summary statistics.** The present review focuses on linear summaries (slopes, means, change scores). Nonlinear summaries such as time to a clinically meaningful threshold or the area under the response-time curve (AUC) may capture treat-

ment effects that linear summaries miss. Efficiency comparisons under realistic data-generating models would be valuable.

2. **Robust variance estimation.** Combining summary statistics with sandwich (HC) standard errors could provide robustness to heteroscedasticity without requiring mixed-model machinery. The properties of such estimators in the summary statistic context are unexplored.
3. **Multiple imputation with summary statistics.** When missing data rates are substantial, computing summary statistics within each multiply imputed dataset and pooling via Rubin's rules could extend the approach to settings where simple complete-case analysis is inadequate.
4. **Software tools.** A dedicated R package implementing the summary measures approach with ANCOVA, baseline adjustment, sample size formulas, and diagnostic plots would lower the barrier to adoption.
5. **Head-to-head regulatory submissions.** Empirical evidence from regulatory submissions where both MMRM and summary statistics were presented as co-primary or sensitivity analyses would illuminate the practical concordance of the two approaches.
6. **Bayesian summary statistics.** Integrating the summary measures approach with Bayesian sample size determination[22] and posterior inference could provide a unified framework for design and analysis.
7. **Extension to non-Gaussian endpoints.** The summary measures framework is well-developed for continuous outcomes. Extension to binary or count outcomes (e.g., proportion of visits with response, total adverse events) remains underexplored.

6 Appendix: Morris et al. (2019) ADEMP compliance audit

This audit was previously nested under the References heading, where it rendered between the References title and the citeproc bibliography (whitepaper M7); it is relocated here as an appendix and re-run against the corrected simulation described in this manuscript rather than reproducing the pre-remediation "Compliant" verdict, which was false: at the time it was written, every MMRM fit had silently failed (M1) and two of three methods were scored against the wrong estimand (M2), so the study did not meet Morris et al.'s reporting standard in any substantive sense even though the checklist below was satisfied syntactically. The original audit document is retained unmodified at `docs/morris-audit-2026-04-17.md` for the historical record; the verdict below supersedes it.

Verdict as of this remediation pass (2026-08-20, second pass): compliant. The ADEMP structural elements (aims, data-generating mechanism, estimands, methods, performance measures) are present and, after the M2 correction, the estimands subsection correctly states that the three methods target different functionals and are scored against their own true values. The ADEMP-recommended null (`\(\delta = 0\)`) condition is now included (M4), MMRM converges in every reported cell rather than failing wholesale (M1), and every reported performance measure carries a Morris Table 6 Monte Carlo SE. The currently committed `sim_09.rds` (and therefore every inline number, table, and figure in this document) is the production-scale run, `n_reps = 1000` across the full 60-condition design grid, so the reported Monte Carlo SEs (about 0.7 percentage points for coverage, about 1.4-1.6 percentage points for power in the least favorable cells) are informative at the precision Morris et al.'s reporting standard presumes. This production run also surfaced a second coding defect not visible in the pilot: `fit_mmrm()` read the t statistic rather than the p-value from `mmrm`'s coefficient table (column 4 instead of column 5), which produced an MMRM Type I error near 0.50 in an initial production-scale attempt. The fix is regression-tested (`inst/tinytest/test_basic.R`) and the run redone; see the `run-sim` chunk in the Methods for detail.

Audit-time gaps and their resolution:

- *Sim chunks `eval=FALSE`* — Resolved. The `run-sim` chunk is `eval=TRUE, cache=TRUE` and runs at render time; cached output is reused on subsequent renders.
- *No `set.seed()` in `sim_study.R`* — Resolved by setting the seed once at the caller side (Morris et al. 2019 §4.1.1) in the `run-sim` chunk: `RNGkind('L'Ecuyer-CMRG'); set.seed(20260310)`.
- *No Monte Carlo SE machinery* — Resolved. The `run_simulation()` function in `sim_study.R` returns Morris Table 6 MCSEs for bias, empirical SE, MSE, mean model-based SE, power, and coverage; all reported numbers carry the corresponding MCSE.
- *No render-blocking sanity checks* — Resolved by `check_simulation_quality()`, added during this remediation pass, which stops the render if any method-condition cell has zero convergence or zero valid replicates, or if SMA-slope coverage under MCAR departs from nominal 0.95 by more than 5 MCSEs plus 0.03 (whitepaper M3, m9).
- *MMRM total fit failure (M1)* — Resolved: the covariance term in `fit_mmrm()` is now the unqualified `us(visit | id)` rather than the unrecognized `mmrm::us(visit | id)`; convergence is verified nonzero (at least 0.998 in every reported cell) in the production-scale run (see the tinytest suite in `inst/tinytest/test_basic.R` for a regression check).
- *Estimand mismatch (M2)* — Resolved in code and text via `truth_for_method()` and the corrected Estimands subsection.
- *No Type I error assessment (M4)* — Resolved: `delta = 0` is now part of the design grid and reported in the Type I error subsection.
- *Production-scale run not yet executed* — Resolved: the committed `sim_09.rds` is the `n_reps = 1000` production run over the full design grid.
- *MMRM p-value read from the wrong `mmrm::summary()` coefficient column (found during the production-scale run)* — Resolved: `fit_mmrm()` now reads `Pr(>|t|)` (column 5) rather than `t value` (column 4); regression-tested in `inst/tinytest/test_basic.R`.

Pilot runs use `N_REPS=100` (or the smaller ad hoc pilot used to verify this remediation) for fast smoke-test rendering; production submission requires `N_REPS=1000` (the default) over the full design grid, as used to generate the results reported in this document.

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